



University of Missouri - Columbia

AIAA Design, Build, Fly Proposal

2025 – 2026



1. Executive Summary

This proposal details the approach of the **Mizzou AeroTigers**, representing the University of Missouri – Columbia, to design, analyze, manufacture, test, and fly a scale aircraft at the 2026 AIAA Design, Build, Fly Competition in Wichita, Kansas. This aircraft must be a *Banner Towing Bush Plane*, capable of conducting a Test Flight, Charter Flight, and Banner Flight, which requires the plane to remotely deploy and release a banner in flight. The aircraft must also possess the ability to carry passengers and cargo that are inserted into the plane during the ground mission. After decomposing mission objectives and performing a sensitivity study on design parameters, a custom optimization program was developed to produce the highest-scoring aircraft. This year's design, **BENGAL**, will be a single-motor aircraft with a high-wing rectangular planform and conventional empennage. It will have a **wingspan of 59 inches**, and it will be capable of simultaneously carrying **12 passengers** and **4 cargo units** for **10 laps**, with **22.2 Wh** of stored energy during Mission 2 (M2). It will also be capable of carrying a deployable banner with a length of **60 inches** and an aspect ratio of 5 for **4 laps** during Mission 3 (M3). To successfully execute this design, a project schedule, manufacturing plan, and testing plan have been developed based on the key milestones of conducting a maiden test flight in December, submitting a holistic design report in February, and attending the Fly-Off in April.

2. Management Summary

2.1 Team Organization: Mizzou AeroTigers is a student-led organization that consists of five main teams: Electronics, Aeronautics, Mechanical, Missions/Software, and Administration. The teams are organized as shown in **Figure 2.1** and each has a team leader responsible for completing tasks on time. The Electronics Team focuses mainly on designing and testing the propulsion, control, and energy-storage systems. The Aeronautics Team focuses on sizing, planform design, and general aerodynamic analysis of the plane. The Mechanical Team focuses on the general design, SolidWorks modeling, and the structures and mechanisms that make up the plane. The Missions/Software Team focuses on maintaining and setting up the flight simulator, developing MATLAB software, and integrating mission objectives into the overall aircraft design. The Administration

Team focuses on event planning and outreach, as well as managing technical documentation. All teams contribute to the design proposal and design report, with two lead engineers supervising project management and scheduling. These leaders are responsible for running general meetings, recruiting new members, acquiring funds, and integrating the work of the five teams.

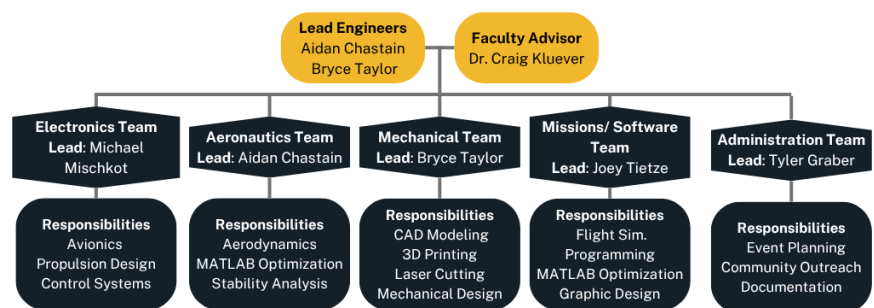


Figure 2.1: Team Organization



2.2 Schedule of Project Milestones: Figure 2.2 presents the AeroTigers' timeline to prepare for the Fly-Off. Maintaining scheduled deadlines is key to developing a high-scoring aircraft with sufficient time for prototyping and testing before the Fly-Off. The AeroTigers' conceptual and preliminary design phases are already complete; Section 3 details these phases and the results. Currently, the detailed design phase is in progress, involving the generation of comprehensive CAD models for manufacturing. Prototype manufacturing is scheduled to begin in early November, aiming for the first test flight to occur in early December. The data and experience gained from each flight will be used to improve the aircraft, with the final design scheduled to be fully manufactured by the end of February. The manufacturing strategy is further detailed in Section 4, while the testing procedures and planned testing schedule are contained in Section 5.

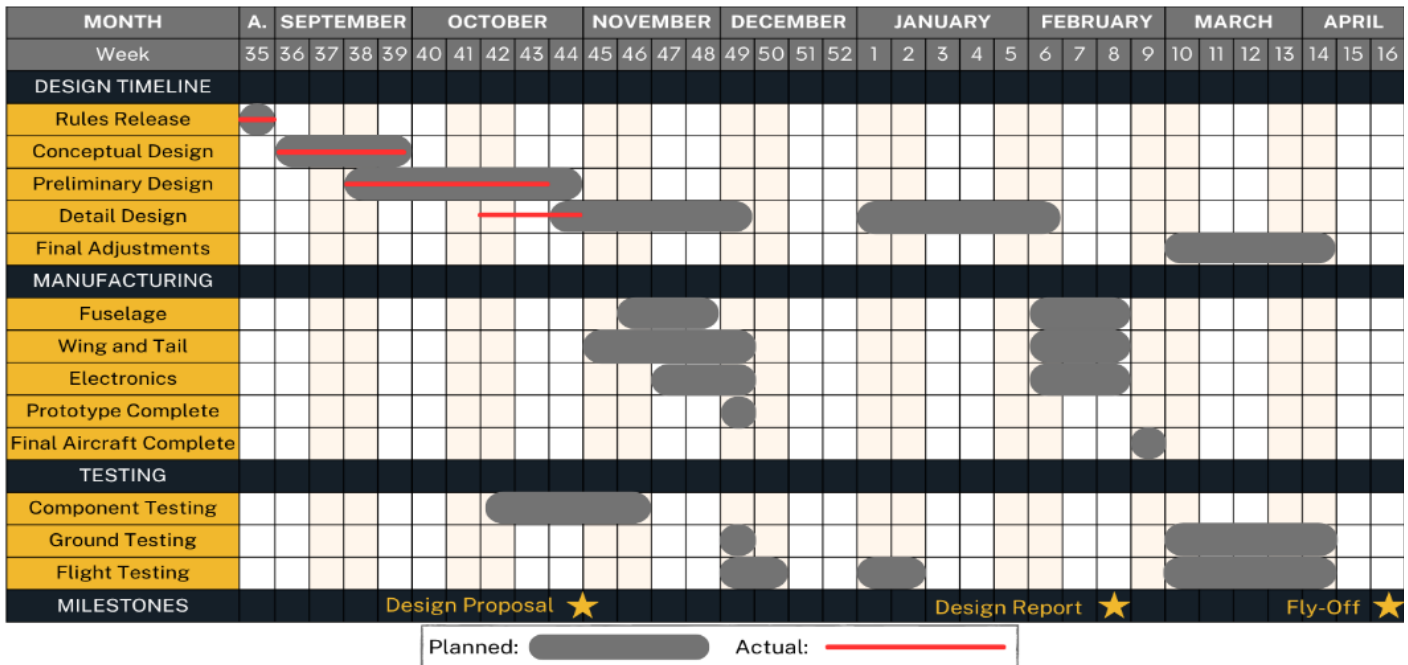


Figure 2.2: Competition Gantt Chart

2.3 Budget: The Mizzou AeroTigers' anticipated budget for the 2026 Fly-Off is detailed in Table 2.1. The anticipated costs will be financed through a combination of fundraising, corporate outreach, and support from the University. This support will cover manufacturing, testing, and anticipated travel costs. This funding will enable the team to bring 10 students, including leadership and key contributors, to the in-person Fly-Off in Wichita, Kansas. The AeroTigers have already acquired some necessary manufacturing tools, including 3D printers, soldering equipment, and laser cutters.

3. Conceptual Design Approach

3.1 Mission Decomposition: This year's challenge of building a banner-towing bush plane will consist of three flight missions and one ground mission. Table 3.1 defines scoring functions and requirements for each mission.

Table 2.1: Budget

CATEGORY	ITEM	ANTICIPATED COST (USD)
Electronics (\$1,120)	Propellers	\$250
	Batteries	\$250
	Servos	\$200
	Electronic Speed Controller	\$220
	Receiver	\$200
Structures (\$2,655)	Carbon Fiber Spars	\$700
	MonoKote Film	\$100
	Landing Gear, Hardware	\$100
	Balsa/Birch Plywood	\$400
	Fasteners/Glues/Tapes	\$100
	PLA Filament	\$55
	Fiberglass Fabric and Resin	\$1,200
Buildroom Supplies and Other (\$800)	Storage Cabinets and Toolchests	\$150
	Miscellaneous Hand Tools	\$100
	Outreach Events	\$250
	Foam Hot-Wire Knife	\$300
Travel (\$2,650)	Lodging (5 nights)	\$1,250
	Gas/Vehicle Rental (Wichita, KS)	\$850
	AIAA Registration Fee	\$550
Total Anticipated Cost		\$7,225
Current Account Balance		\$5,739
Expected Contributions		\$6,500

3.2 Trade Studies: Trade studies were conducted using the decision matrices detailed in **Table 3.2**. Quantitative data were gathered and analyzed to select the banner material, concluding that Monolite ripstop provides the best balance of weight, efficiency, and aerodynamic effects for the banner. Key aircraft configuration parameters (wing planform, banner release mechanism, and tail configuration) were rated using a weighted 1-10 scale with 10 being the best score possible.

Table 3.1: Mission Requirements Breakdown

MISSION	SCORING	REQUIREMENTS	SUB-SYSTEM REQUIREMENTS
M1: Test Flight	$M_1 = 1.0$	- Prepare for flight in under 5 min. - Complete 3 laps in under 5 min.	- Aircraft wingspan shall not exceed 5 ft. - Aircraft shall sustain controlled and stable flight in varying weather conditions. - Aircraft weight shall not exceed 55 lbs.
M2: Charter Flight	$M_2 = 1.0 + \frac{[Income - Cost]_{MU}}{[Income - Cost]_{Max}}$ $Income = P(6 + 2L) + C(10 + 8L)$ $Cost = \frac{LB_{cwp}(10 + 0.5P + 2C)}{100}$	- Prepare aircraft and load passengers and cargo in under 5 min. - Complete as many laps as possible in 5 min. - Aircraft shall carry a minimum of 1 cargo and 3 passengers. - Aircraft shall carry a minimum of 3 passengers for each cargo. - Aircraft shall minimize propulsion battery capacity.	- Compartments shall be optimized to maximize passenger and cargo capacity. - Passenger compartment shall be built parallel to horizontal plane. - Airframe shall minimize weight and drag to enhance flight efficiency. - Compartments shall securely restrain cargo and passengers to prevent movement during flight.
M3: Banner Flight	$M_3 = 2.0 + \frac{[L \cdot B_{len}/RAC]_{MU}}{[L \cdot B_{len}/RAC]_{Max}}$ $RAC = 0.05b + 0.75$	- Prepare aircraft and load banner in under 5 min. - Complete as many laps as possible in 5 min. - Banner shall have a minimum length of 10 in. - Banner shall have a maximum aspect ratio of 5. - Banner shall be deployed after 1st upward turn and released upon crossing the finish line in final lap. - Aircraft shall minimize wingspan and maximize banner length.	- Banner stowing mechanism shall ensure banner stays in vertical orientation during flight. - Banner stowing mechanism shall secure the banner externally without interfering with flight functions. - Banner stowing, deployment, and release mechanism shall be built to minimize drag and ensure structural integrity. - Banner release and deployment mechanisms shall operate reliably at high airspeeds.
Ground Mission	$GM = \frac{t_{min}}{t_{MU}}$	- Aircraft shall enable loading and unloading of passengers and cargo as quickly as possible. - Passengers shall be loaded individually. - Aircraft shall enable rapid installation, deployment, and release of the banner.	- Passenger and cargo compartments shall be designed for efficient removal and insertion. - Banner storage system shall be designed for efficient insertion. - Banner release and deployment systems shall be designed for efficient execution.

A rectangular wing allows for ease of manufacturing with minor sacrifices in aerodynamic performance. A servo-latch release allows for consistent and reliable banner release without adding significant weight, and a conventional empennage provides the best manufacturability while maintaining excellent aerodynamics.

Table 3.2: Decision Matrices

Banner Material					Wing Planform					
	Scoring Weight	Monolite Ripstop	Rip-Stop Nylon	Dyneema Composite 0.67		Scoring Weight	Elliptical	Rectangular	Tapered	Swept
Weight	30%	8.7	3.7	10	Manufacturability	40%	2	10	6	3
Air Permiability	20%	9	8	6	Weight	15%	10	8	9	8
Surface Texture/ Flexibility	20%	9	8	6	Structural Integrity	30%	5	10	7	6
Drag/ Flutter Stability	30%	8	7.2	6.9	Aerodynamic Performance	15%	10	8	9	8
Weighted Result	100%	8.61	6.47	7.47	Weighted Result	100%	5.3	9.4	7.2	5.4
Banner Release Mechanism					Tail Configuration					
	Scoring Weight	Spring Loaded Ejection	Servo-Latch Release	Solenoid Ejection		Scoring Weight	T-Tail	V-Tail	Conventional Tail	U-Tail
Manufacturability	25%	3	8	5	Manufacturability	40%	6	7	10	7
Weight	15%	4	8	8	Structural Integrity	20%	3	7	7	10
Simplicity	20%	4	7	4	Weight	10%	8	10	9	6
Reliability	40%	7	8	9	Aerodynamic Performance	30%	10	10	8	10
Weighted Result	100%	4.95	7.8	6.85	Weighted Result	100%	6.8	8.2	8.7	8.4

3.3 Sensitivity Analysis: A sensitivity study was conducted using MATLAB to analyze the impact of key scoring and design variables.

Figure 3.1 illustrates how each mission objective impacts the scoring function. The results indicate that the number of laps completed during M3, and M3 banner length have the largest impact on score, with M2 laps, cargo and passenger count having a slightly smaller effect. GM time has an inverse, nonlinear relationship with score. It is also clear that battery capacity must be minimized to complete five minutes of flight, as excesses in battery capacity have a negative impact on the overall score. Wingspan impacts the score through the rated aircraft cost (RAC) and has a relatively small impact on the overall score. **Figure 3.2** depicts how wing aspect ratio and M3 banner length impact the overall score at a constant thrust value. The region above the curve represents infeasible configurations due to excessive drag. Larger aspect ratios decrease the maximum takeoff weight, along with the ability to tow a larger banner, but also decrease induced drag, allowing for higher M2 flight speed. The results from the sensitivity study were used as inputs for generating optimized sizing in the preliminary design. The overall process is illustrated in **Figure 3.3**.

3.4 Preliminary Design:

3.4.1 Aircraft Design

Preliminary design of the aircraft is an iterative process that focuses on three distinct categories: sensitivity analysis, optimization code, and stability analysis. Following the mission decomposition and sensitivity studies, a MATLAB optimization code, titled TigerOptimization, utilized MATLAB's Global Optimization Toolbox to obtain optimal parameters. TigerOptimization calls the function "gamultiobj", to access a genetic algorithm that obtains the best combination of parameters to maximize score for the missions as identified in Section 3.1. TigerOptimization outputs possible solutions with maximized scores, and a decision matrix is made to select the best set of parameters. **Table 3.2** shows the decision matrix utilized.

A blended rectangular fuselage was selected to achieve a high fineness ratio, minimizing induced drag while ensuring room for passenger and cargo bays. The wingspan was chosen to be **59 inches** to minimize the effect of expected perturbances, with sensitivity analysis revealing the relatively low impact of rated aircraft cost. Preliminary stability analysis has placed an angle of incidence of 1.5° on the wing, and -1° on the tail. In addition, it has located the neutral point of the aircraft at

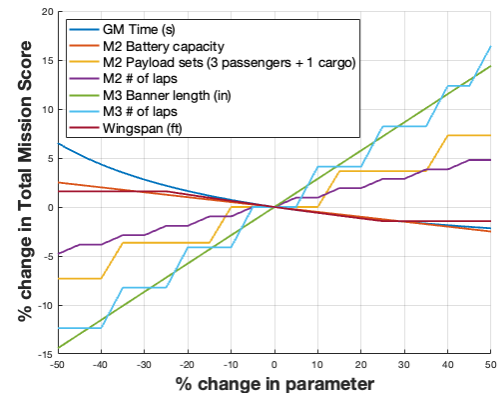


Figure 3.1: Sensitivity of Scoring Variables

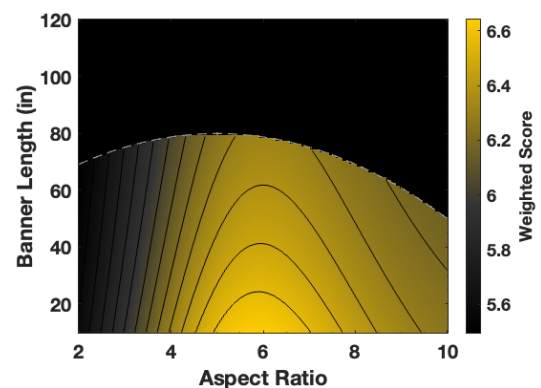


Figure 3.2: Analysis of Design Parameters

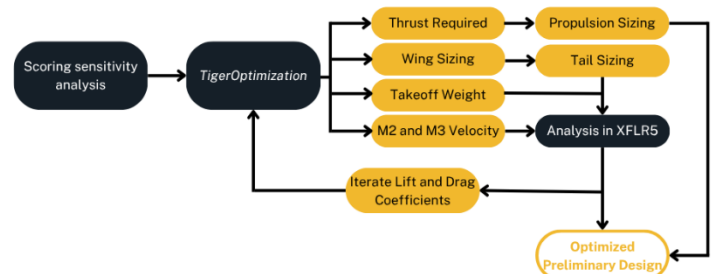


Figure 3.3: Preliminary Design Flowchart

50% MAC and has placed the CG at **26% MAC** to create a static margin of **24% MAC**. The M2 payload will be located such that this CG is maintained, while ballast will be used to achieve the same CG during M3. The battery, ESC, and motor will all be placed in the foremost bay.

3.4.2 Cargo and Passenger Storage

Accessibility and balanced weight distribution were the primary considerations when designing the cargo and passenger storage systems. The layout and orientation of these components are shown in **Figure 3.4**. Passengers are secured in a carrier located at the rear of the fuselage, mounted to the top panel, with a hatch above for quick and easy access. Cargo is housed in a sleeve positioned between two firewalls, located at a position determined by preliminary analysis, minimizing its impact on overall flight stability. A side hatch will be used to insert the sleeve into the fuselage.

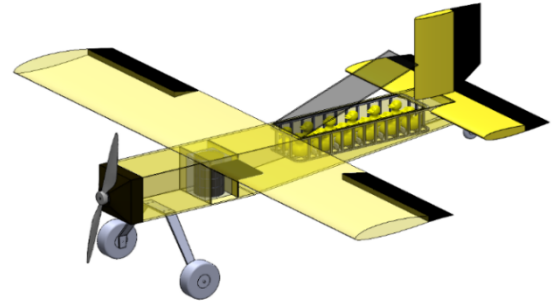


Figure 3.4: Aircraft CAD Model in M2 Configuration

3.4.3 Banner Design and Release Mechanism

The banner will be stowed externally, opposite the cargo hatch, to ensure ground clearance during takeoff and to allow room for deployment mechanism electronics.

A servo-latch style banner release mechanism will be used, as illustrated in **Figure 3.5**. The banner is stored in the rolled position and secured to an outer shell using an elastic chord with a hook on one end. The hook is placed through a servo latching mechanism, which, upon servo actuation, releases the chord, dropping the banner from the housing. Prior to landing, an actuated pin release system will detach the banner assembly from the aircraft in flight.

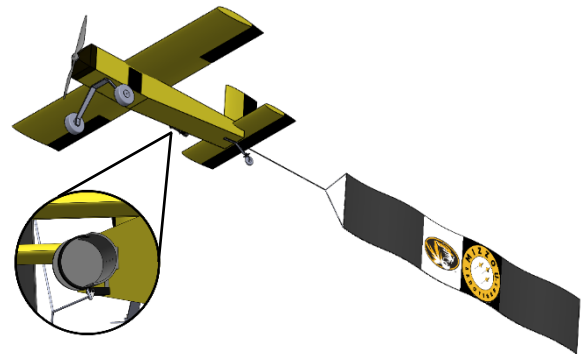


Figure 3.5: Aircraft CAD Model in M3 Configuration

3.4.4 Propulsion

The main design consideration for the propulsion system is the capability of providing adequate thrust for the optimal payload while still drawing low levels of electric current. Additional design considerations included cost, and the ability to reuse components from previous competition aircraft. Thus, the **Scorpion Power Systems SII-4025-520kV** motor was selected. The motor will be placed in a tractor configuration for simplicity. This motor is a **6S** brushless motor capable of spinning a **17 x 10E** propeller and producing **12 lbf** of thrust. The motor draws 80 A at takeoff, but only 8 A while cruising, making it suitable for this competition in which lower battery capacity improves scoring. Assuming a 10 second ground roll and a 4 minute and 50 second cruise, the propulsion battery will be a **6S 1000 mAh LiPO**, with a buffer of 133 mAh for extraneous maneuvers necessary at higher speeds. The electronic speed controller will be the **Spektrum Avian 100 Amp Brushless Smart ESC**.

3.5 Mission Targets: When optimizing aircraft design and configuration, mission targets are the primary concern. TigerOptimization was used to calculate the overall score of each solution presented. For Mission 1, **3 laps** will be flown within 150 seconds, given a 10 second takeoff and drag optimized cruise speed. For Mission 2, **10 laps** will be flown with **4 cargo units, 12 passengers**, and a **6S 1000 mAh battery**. For Mission 3, **4 laps** will be flown with a banner length of **60 inches**, and a banner height of 12 inches.

4. Manufacturing Plan

Figure 4.1 outlines the manufacturing flow. Each section of the aircraft will be broken down into individual components and modeled in SolidWorks. The primary framework, including ribs, formers, and fuselage sides, will be cut from 1/8-inch and 1/16-inch sheets of balsa and birch wood using a 60W CO2 laser cutter. More complex components will be 3D printed, using lightweight PLA for non-structural components and PAHT-CF for structural components, with optimized infill patterns to maximize strength while minimizing weight. Key structural areas, such as wing box and integrated tail, will be reinforced with carbon fiber tubes to withstand flight loads. After manufacturing each subsystem, the aircraft will be assembled and sealed with a MonoKote exterior skin. The final competition aircraft will be designed and assembled in the same way with improvements made to the initial design from experience and testing.

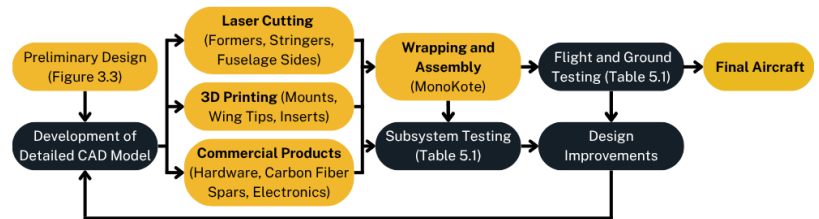


Figure 4.1: Manufacturing Flowchart

5. Testing Plan

The testing plan for this aircraft is structured to validate performance and reliability across all three missions. Each test detailed in **Table 5.1** targets a critical requirement while providing early data to reduce design risk before competition. Testing begins with component-level evaluations to confirm subsystem functionality, followed by progressive ground and flight trials that confirm mission readiness. Results will drive iterative improvements with the goal of delivering a fully compliant, flight-proven aircraft by April 2026.

Table 5.1: Testing Plan

CATEGORY	TEST	METHOD	GOAL	DUE DATE
Component Testing	Static Thrust Test	Measure thrust, battery usage, and power on a test stand	Select propeller/battery and create thrust model	10/16/25
	Electronics Bench Test	Attach all electronic systems on test bench	Ensure working avionics and propulsion subsystems	10/31/25
	Banner Operation Test	Test banner deployment and release mechanism	Ensure reliability of banner deployment and release	11/10/25
	Payload Security Test	Subject payload to six-axis motion and observe payload response	Ensure cargo and passenger security in all flight conditions	11/10/25
	Banner Drag Test	Measure banner drag using spring scale attached to moving vehicle	Estimate banner parasite drag at M3 optimized velocity	11/10/25
Ground Testing	Landing Gear Test	Drop loaded aircraft onto landing gear	Ensure reliability of landing gear	12/02/25
	Wing Tip Test	Hold each mission-configured aircraft by its wingtips	Ensure structural stability and locate static margins	12/02/25
	Assembly Test	Test GM speed and consistency of installing passengers and cargo, and stowing banner	Select ground crew and optimize GM score	12/05/25
	Taxi Test	Test movement of aircraft on ground	Ensure landing gear functionality, balance, and steering capability	12/06/25
	Technical Inspection	Perform thorough inspections of the aircraft's construction	Demonstrate ability to pass technical inspection	12/06/25
Flight Testing	Takeoff Distance Test	Measure takeoff distance for different weights	Determine takeoff distance to assess possible hazards associated with long ground roll	12/07/25
	Flight Endurance Test	Measure battery usage in-flight with M2 configuration	Identify maximum number of laps with optimized battery capacity	12/14/25
	Mock Flight Missions	Practice each flight mission	Ensure stability, optimize velocity, maximize payload	01/11/25